Skin Dataset Appendix

1.1: Dataset Overview

This dataset contains dermoscopic images of skin lesions collected and curated by the International Skin Imaging Collaboration (ISIC) as part of a data challenge called the MILK10k Challenge from ISIC. Thus, the dataset is commonly used for machine learning.

The dataset itself is a collection of contributions from multiple international clinical centers, ensuring diversity in imaging devices, patient populations, and lesion types.

Total observations analyzed: **10,480 dermoscopic images but 5,240 unique lesions**

Each oberservation corresponds to a paired set of images:

- A dermoscopic (magnified) image
- A clinical close-up photograph of the same lesion

*Accompanying each pair is the metadata that describes the patient and lesion characteristics, that is used in the machine learning

Class Breakdown of Lesion Diagnostics (11 Categories):

1. Actinic keratosis / intraepidermal carcinoma (AKIEC)
2. Basal cell carcinoma (BCC)
3. Other benign proliferations, including collision tumors (BEN_OTH)
4. Benign keratinocytic lesion (BKL)
5. Dermatofibroma (DF)
6. Inflammatory and infectious conditions (INF)
7. Other malignant proliferations, including collision tumors (MAL_OTH)
8. Melanoma (MEL)
9. Melanocytic nevus (NV)
10. Squamous cell carcinoma / keratoacanthoma (SCCKA)
11. Vascular lesions and hemorrhage (VASC)

1.2: Unit of Observation

- **Unit of Observation:** A single dermoscopic image file.
- **Unit of Analysis:** The unique skin lesion (lesion_id).

- **Data Integrity Protocol:** To prevent Data Leakage, the train/test split was performed at the lesion_id level. This ensures that if one image of a lesion is in the test set, the second image of that same lesion cannot appear in the training set, maintaining the validity of the statistical significance tests.

1.3: Variable Documentation

1. Variable: image_id

- Type: String
- Description: Unique identifier for each image file.
- Example Value: IL_0000652

2. Variable: image_type

- Type: Categorical (String)
- Description: Specifies modality of the image.
- Two types: clinical: close-up and dermoscopic

3. Variable: ISIC_id

- Type: String
- Description: Unique identifier linking previously existing dermoscopic images if and when applicable.

4. Variable: lesion_id

- Type: String
- Description: Unique identifier for each lesion (associated with two images)

5. Variable: age_approx

- Type: Integer
- Description: Patient age approximation grouped into 5-year intervals.

6. Variable: Sex

- Type: Categorical (string)
- Description: Biological sex of the patient.

7. Variable: skin_tone_class

- Type: Ordinal
- Description: Skin tone classification.
- Range from 0 being very dark and 5 being very light.

8. Variable: site

- Type: Categorical (string)
- Description:
- Anatomical location of the lesion.

9-15. Variable: MONET_(feature annotations)

- Type: Continuous (probabilities)
- Description: Probabilistic annotations of dermoscopic features generated via the MONET framework.
- Feature categories include:
    - Ulceration / Crust
    - Hair
    - Vasculature
    - Erythema
    - Pigmentation
    - Dermoscopy Liquid Artifacts
    - Skin markings/ink

1.4: Figures and Visualization

1. Distribution of Image_Type, Sex, Skin Tone Class, Site for Lesions

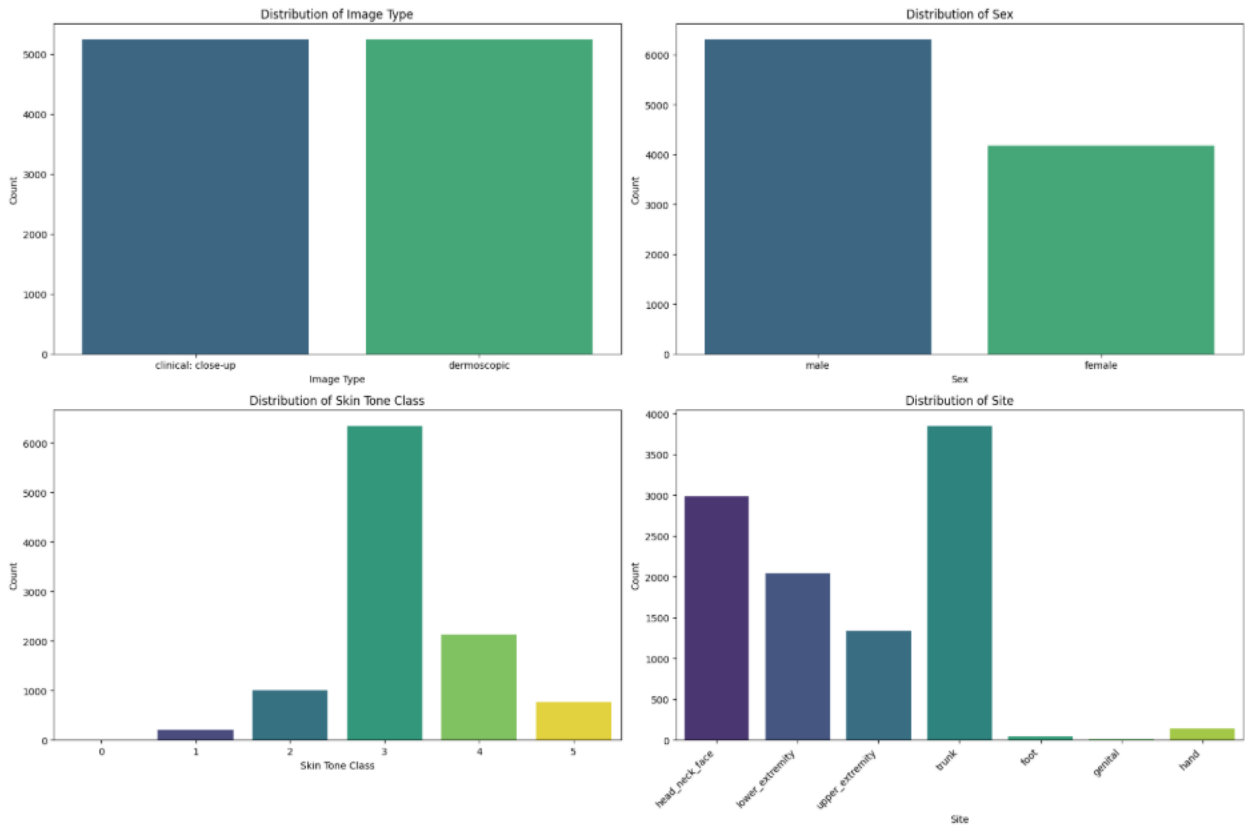


Figure Description: Four different distributions of patient traits/lesion information.

Variable Used:

- Image_Type
- Sex
- Skin_Tone_Class
- Site

Interpretation:

Distribution of Image Type- There appears to be an even distribution of image type represented which makes sense because every image should be paired so this visualization simply emphasizes that both are dually represented.

Distribution of Sex- Males are more represented in the data than females in the lesion data.

Distributions of Skin Tone Class- There is a range of skin tones represented, but there is a lack of darker skin tones represented which is a common limitation within dermatology where darker skin tones are not represented which makes diagnosis not as equitable.

Distribution of Site- There are a variety of sites represented for where lesions appear, but foot, genital, and hand have a lot less which may make diagnosis of those areas more difficult due to the rarity.

2. Distribution of Lesion Types

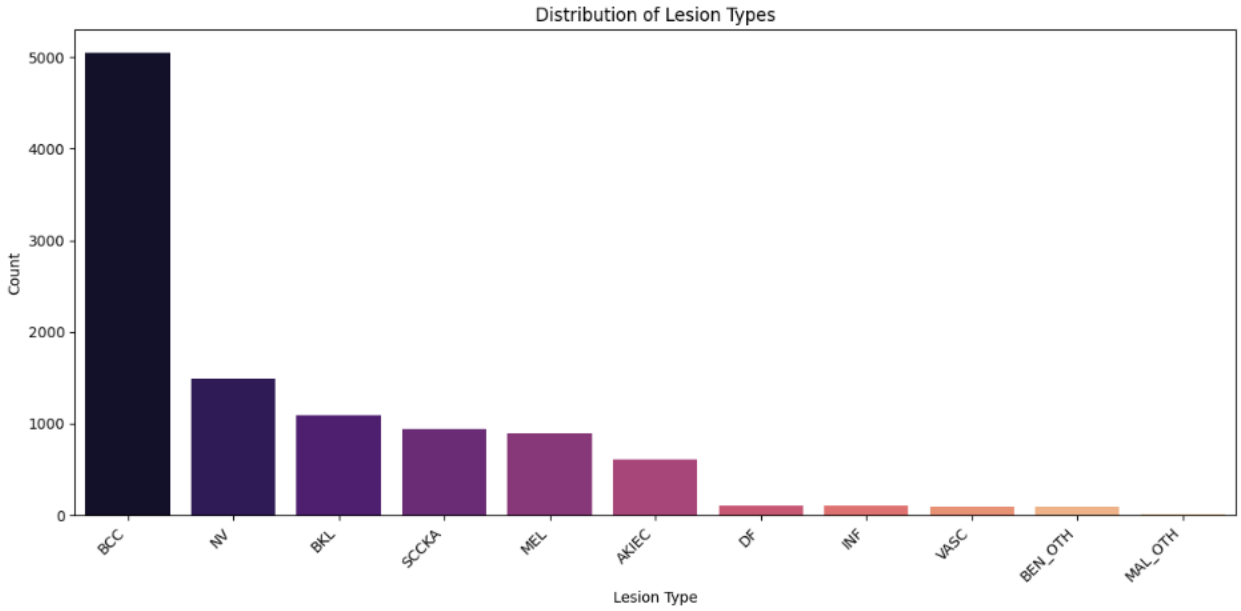


Figure Description: This figure breaks down the number of each lesion type within the data set

Variable Used:

- Lesion Type

Interpretation: The dataset exhibits extreme class imbalance. Basal Cell Carcinoma (BCC) represents nearly 50% of the observations, while others like MAL_OTH represent less than 0.1%. This was addressed through weighted sampling to prevent the model from defaulting to the majority class.

3. 11-Class Image-Only vs. Multimodal F1 Scores per Diagnosis

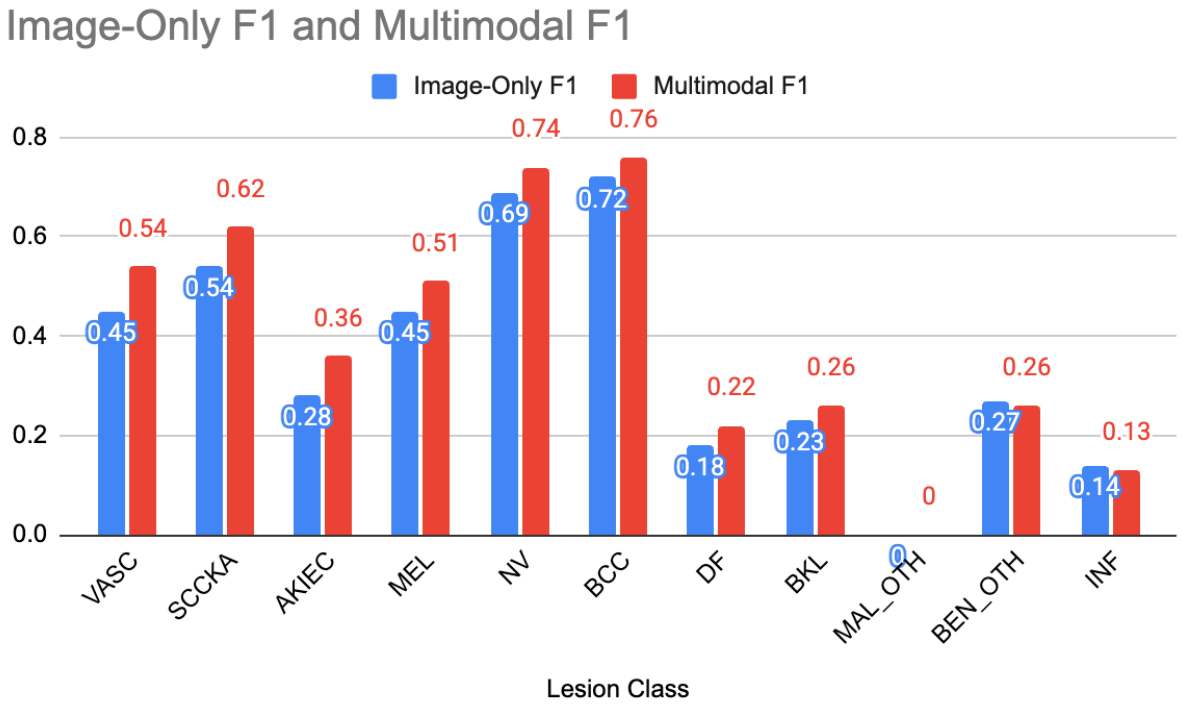


Figure Description: This figure shows the F1-scores of each individual lesion type

Variables Used:

- F1-Score
- Diagnosis

Interpretation: The Multimodal model improved F1-scores for 8 out of 11 categories. High-stakes diagnostic categories such as SCCKA (+0.08) and AKIEC (+0.08) saw significant improvements. Small performance drops in BEN_OTH and INF are likely due to extremely low support ($N < 20$) and the lack of distinct clinical metadata markers for those specific pathologies.

4. Binary Classification Confusion Matrices (Image-Only vs. Multimodal)

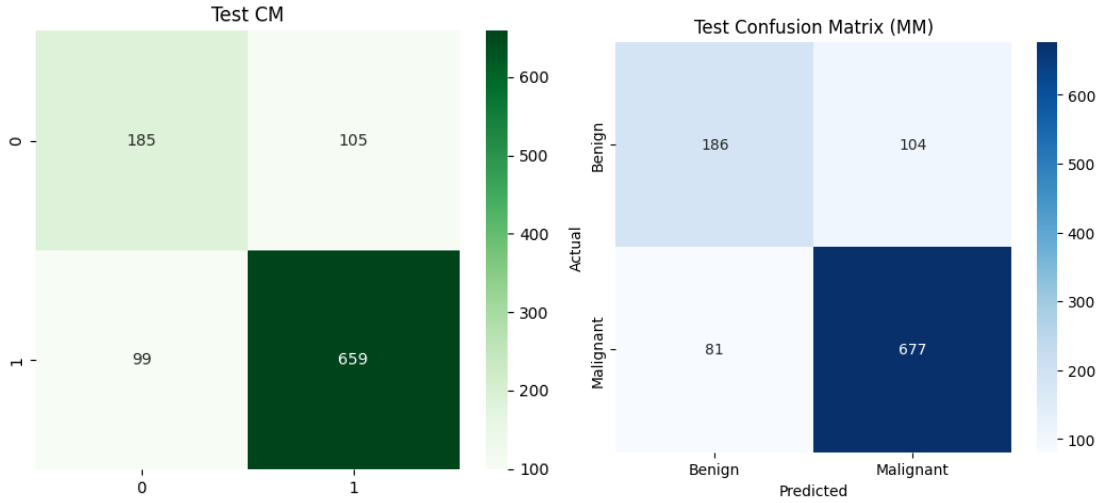


Figure Description: These two confusion matrices show the results of using the respective binary models on the test data

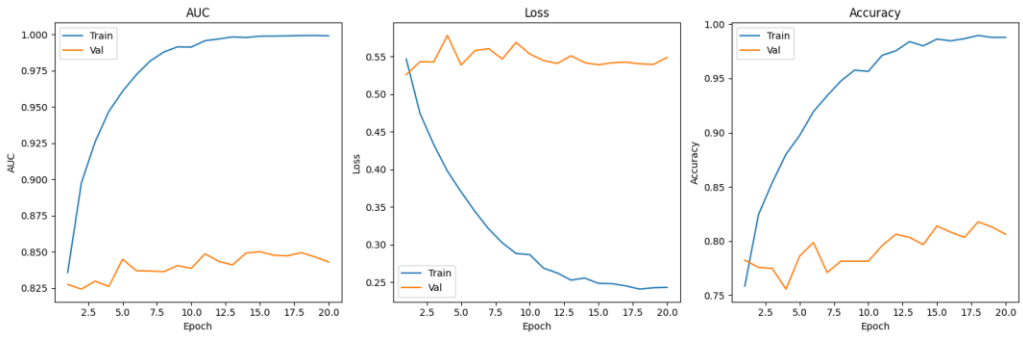
Variable Used:

- Predicted (0 or 1)
- Actual (0 or 1)

Interpretation: The multimodal integration primarily succeeded in reducing False Negatives in malignant cases. This shift suggests that clinical metadata provides the necessary context to identify malignant lesions that may appear visually ambiguous.

5. Binary vs. 11-Class Learning Plots (Multimodal)

Binary



11-Class

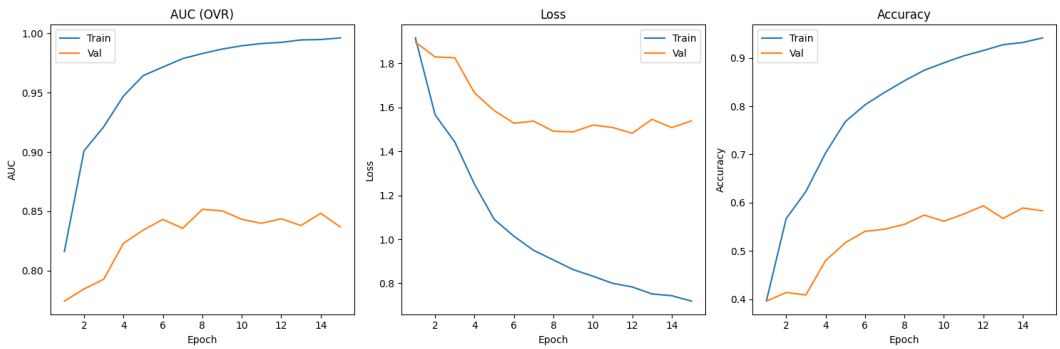


Figure Description: These figures show the changes in validation results as the models progress through their epochs.

Variables Used:

- AUC
- Accuracy
- Loss
- Epoch Number

Interpretation: The 11-class model utilized fewer epochs because its complex decision boundaries led to faster overfitting. It exhibited rapid initial convergence with sharp "jumps" in learning, followed by minor fluctuations during granular fine-tuning. Conversely, the binary model followed a more gradual upward trajectory but experienced larger early-stage fluctuations, reflecting the challenge of establishing a singular threshold for a diverse "Malignant" category.

1.5: Statistical Procedures

Performance was measured using **Accuracy** and **One-vs-Rest AUC**. Comparison between the Image-Only and Multimodal architectures was conducted using:

- **Two-Proportion Z-Test:** To evaluate the significance of Accuracy deltas.
- **Hanley & McNeil Method:** To evaluate the significance of AUC shifts.
- **Significance Threshold:** $\alpha = 0.05$

Experiment	Metric	Image-Only	Multimodal	Difference	P-Value	Significance
Binary	Accuracy	80.53%	82.35%	+1.82%	0.2839	Not Significant
Binary	AUC	0.8247	0.8489	+0.0242	0.1337	Not Significant
11-Class	Accuracy	57.32%	62.66%	+5.34%	0.0022	Highly Significant
11-Class	AUC	0.8055	0.8330	+0.0275	0.0450	Significant